An almost-all non-halting theorem for a Busy-Beaver renewal map

BB(6) research program

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Abstract

A holdout of the $\Sigma(6)$ Busy-Beaver search reduces, on its natural gait, to an explicit piecewise-affine renewal map $\mathbf{R}$ on $\mathbb{Z}^2_{\geq 1}$ that halts exactly when its orbit enters a fatal set $\mathcal{D}$. We prove that the seeds whose $\mathbf{R}$-orbit ever meets $\mathcal{D}$ form a set of natural density zero: almost every seed yields a non-halting computation. The argument identifies the first-return map of $\mathbf{R}$ to a distinguished section with a countable full-branch piecewise-Möbius Markov map G of the interval, whose branch endpoints obey the explicit recurrence $\rho_{k+1} = (3\rho_k + 1)^{-1}$ and accumulate at the algebraic number $\rho^* = (\sqrt{13} - 1)/6$. We show G is eventually uniformly expanding and of summable variation; Rychlik's theorem then supplies an absolutely continuous invariant measure with exponential decay of correlations, and a Borel–Cantelli estimate on a geometrically summable halting hazard yields the density-zero conclusion. The continuum model is treated completely, with each ingredient verified against the exact dynamics; the transfer to the integer map and to natural density is reduced to two standard estimates (persistence of the spectral gap under an $O(1/a)$ branch-tail truncation, and Jordan-measurability of the halting set), isolated explicitly in §7 and §9. The result is an almost-all theorem in the sense of Tao's Collatz theorem; it is silent on any single seed, and the halting of the certified seed $(6, 13)$ remains Collatz-hard.

1 Introduction

The Busy-Beaver function $\Sigma(6)$ is not known, and its determination is obstructed by a small number of *holdout* machines whose halting is equivalent to an open arithmetic conjecture. One such machine, on its natural macroscopic gait, is governed by an integer renewal map. Its per-orbit halting is Collatz-hard; the present paper proves the corresponding *almost-all* statement, in the spirit of Tao's theorem that almost every Collatz orbit attains almost bounded values [5].

Definition 1. For $(a, b) \in \mathbb{Z}^2_{\geq 1}$ put $\beta(a) = a - 1$ if a is odd and $\beta(a) = a - 2$ if a is even, and $(\alpha, \beta) = (6, a - 1)$ or $(9, a - 2)$ accordingly. The renewal map is

$$\mathbf{R}(a, b) = \begin{cases} (a + 6,\ b - 3), & b \text{ even}, \qquad (T) \\ (\alpha + 3(b - 3),\ \beta - (b - 3)), & b \text{ odd},\ b < \beta, (C) \\ (\alpha + 3\beta + 3,\ b - \beta - 4), & b \text{ odd},\ b > \beta. (F) \end{cases}$$

The fatal set is $\mathcal{D} = \{b \leq 2\} \cup \{b = 4\} \cup \{b \text{ odd}, b > \beta, b - \beta \in \{1, 3, 5\}\}$. The machine halts on the orbit of (a, b) iff that orbit meets $\mathcal{D}$.

Theorem 1. $\lim_{N\to\infty} \dfrac{\#\{(a, b) : a, b \leq N,\ \textit{the } \mathbf{R}\textit{-orbit of } (a, b) \textit{ meets } \mathcal{D}\}}{N^2} = 0$. *That is, all but a density-zero set of seeds are non-halting.*

Our strategy is to pass to a first-return map, to prove that in the correct coordinate it is a classical expanding Markov map of the interval, and to control the halting hazard. The linear parts of T, C, F are

$$T_\ell = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right), \qquad C_\ell = \left(\begin{smallmatrix} 0 & 3 \\ 1 & -1 \end{smallmatrix}\right), \qquad F_\ell = \left(\begin{smallmatrix} 3 & 0 \\ -1 & 1 \end{smallmatrix}\right), \tag{1}$$

with determinants $+1, -3, +3$; the factor 3 is the Collatz multiplier and T_ℓ is parabolic.

2 The parity grammar

Lemma 1. *Every orbit's symbol sequence satisfies $T \to \{C, F\}$, $C \to T$, $F \to \{C, F\}$. In particular TT never occurs.*

Proof. $\beta(a)$ is even for all a. A T-step needs b even and outputs $b - 3$, odd; hence the next step is C or F. A C-step has b odd and outputs $\beta - (b-3) = \beta - b + 3$, which is even $-$ odd $+$ odd $=$ even; hence the next step is T. An F-step has b odd and outputs $b - \beta - 4 =$ odd $-$ even $-$ even $=$ odd; hence the next step is C or F. These are identities in the parities of the integers involved. □

Corollary 1. *The parabolic generator is never iterated: between two consecutive F-steps the orbit reads a word $(CT)^k F$ for some $k \geq 0$. Consequently* **R** *exhibits no Pomeau–Manneville intermittency.*

3 The continuum model of the first-return map

Let $S = \{(a, b) : b > \beta(a),\ b \text{ odd}\}$ be the F-section. By Corollary 1 the first-return map $\mathbf{R}_S : S \to S$ (defined off the halting orbits) exists, and a return with k intervening CT-blocks applies F then $(CT)^k$. Since $T_\ell = I$, the scale-asymptotic linear part of that return is

$$P_k = C_\ell^{\,k} F_\ell = \left(\begin{smallmatrix} p_k & q_k \\ r_k & s_k \end{smallmatrix}\right), \qquad |\det P_k| = 3^{k+1}. \tag{2}$$

We coordinatize S by the scale-invariant ratio $\xi = b/a$. Dropping the bounded translations (justified in §7), the generator actions on ξ are the Möbius maps

$$F:\ \xi \mapsto \frac{\xi - 1}{3} \quad (\xi > 1), \qquad C:\ \xi \mapsto \frac{1 - \xi}{3\xi} \quad (\xi < 1), \qquad T:\ \xi \mapsto \xi. \tag{3}$$

Indeed $F_\ell(a, b) = (3a, b - a)$ gives $\xi \mapsto (b - a)/(3a) = (\xi - 1)/3$, and $C_\ell(a, b) = (3b, a - b)$ gives $\xi \mapsto (a - b)/(3b) = (1 - \xi)/(3\xi)$. Write $c(x) = \dfrac{1 - x}{3x}$ for the C-map on $(0, 1)$; it is a decreasing bijection $(0, 1) \to (0, \infty)$ with

$$c^{-1}(y) = \frac{1}{3y + 1}, \qquad c(x) > 1 \iff x < \tfrac{1}{4}. \tag{4}$$

Definition 2. Let $\rho = \dfrac{\xi - 1}{3} \in (0, \infty)$ be the post-F ratio. The *induced map* G on $(0, \infty)$ is the first-return of the continuum dynamics (3) to the post-F line: from ρ_0, if $\rho_0 > 1$ set $k = 0$ and return; else apply c repeatedly, letting k be the number of applications until the value first exceeds 1.

4 The branch partition is an explicit Markov partition

Lemma 2. *Define $\rho_0 = 1$ and $\rho_{k+1} = (3\rho_k + 1)^{-1}$. Then $\rho_k = |s_k/r_k|$ with (r_k, s_k) the bottom row of (2); the first values are $1, \frac{1}{4}, \frac{4}{7}, \frac{7}{19}, \frac{19}{40}, \frac{40}{97}, \ldots$; the subsequences of even and odd index decrease and increase respectively to*

$$\rho^* = \lim_k \rho_k = \frac{\sqrt{13} - 1}{6}, \quad \text{the positive root of} \quad 3\rho^2 + \rho - 1 = 0. \tag{5}$$

The branch of G carrying exactly k blocks has domain $I_k = (\rho'_{k+1}, \rho'_{k-1}) \cap (0, \infty)$ where $\{\rho'_j\}$ is the monotone reordering of $\{\rho_j\}$; concretely $I_0 = (1, \infty)$, $I_1 = (0, \frac{1}{4})$, and for $k \geq 2$, I_k is the interval between the two consecutive endpoints ρ_k, ρ_{k-2} on the same side of ρ^*.

Proof. The map c sends $(0, \frac{1}{4})$ onto $(1, \infty)$ and $(\frac{1}{4}, 1)$ onto $(0, 1)$, by (4). Hence starting from $\rho_0 \in (0, 1)$ the number k of c-iterates needed to first exceed 1 is governed by the backward orbit of the threshold 1 under c: the value exits after exactly one further step iff it lies below $\frac{1}{4} = c^{-1}(1)$, after two iff it lies in $c^{-1}((0, \frac{1}{4})) = (c^{-1}(\frac{1}{4}), 1)$, and so on. Thus the endpoints are the backward images $c^{-k}(1)$; since $c^{-1}(y) = (3y + 1)^{-1}$, they satisfy $\rho_{k+1} = (3\rho_k + 1)^{-1}$ with $\rho_0 = 1$, giving the displayed values. The recurrence is a Möbius iteration $\rho \mapsto (3\rho+1)^{-1}$ whose unique attracting fixed point solves $\rho(3\rho + 1) = 1$, i.e. (5); the derivative $-3/(3\rho + 1)^2$ at ρ^* has modulus $3\rho^{*2} < 1$, and its negative sign gives the claimed even/odd interleaving and monotone convergence. Finally, $|s_k/r_k|$ obeys the same first-order recurrence: from $P_{k+1} = C_\ell P_k$ one gets $r_{k+1} = p_k - r_k = 3r_{k-1} - r_k$ and likewise for s_k, whence $-s_{k+1}/r_{k+1} = (3s_{k-1} - s_k)/(r_k - 3r_{k-1})$; a one-line computation using $-s_k/r_k = \rho_k$ reduces this to $(3\rho_k + 1)^{-1}$, so $|s_k/r_k| = \rho_k$ by induction from $|s_0/r_0| = 1$. □

Proposition 1. *On each branch I_k, G equals the Möbius map of P_k and is a homeomorphism of I_k onto $(0, \infty)$. Thus G is a countable full-branch piecewise-Möbius Markov map of the half-line, with the single accumulation point ρ^*; equivalently, after the standard compactification $z = 1 - 1/\xi$ (so $\rho = z/(3(1 - z))$), a full-branch Markov map of $[0, 1]$.*

Proof. On I_k the return is $F \circ (CT)^k$, whose linear part is $P_k = C_\ell^k F_\ell$; a composition of Möbius maps is Möbius, so $G|_{I_k}$ is the Möbius map of P_k. It is strictly monotone (nonzero derivative, (6) below) and, by Lemma 2, carries the two endpoints of I_k to 0 and ∞ (the endpoints are precisely where the itinerary changes, i.e. where the Möbius image hits the section boundary); monotonicity then forces the image to be all of $(0, \infty)$. Fullness on every branch is the Markov property. □

Remark 1. The branch index k is a genuine function of ρ *for the model* G, being the first-exit time of the deterministic itinerary of Definition 2. For the exact integer map at finite scale a the partition is only approximate: near each endpoint ρ_k the assignment of k can differ from the model on a set of ρ-width $O(1/a)$ (the translations of Definition 1 perturb the itinerary boundary). Thus the clean Markov structure is a property of the model, and the finite-a boundary discrepancy is part of the perturbation controlled in §7; it is *not* a hidden skew product, since $z = 1 - 1/\xi$ and $\rho = (\xi - 1)/3$ are monotone reparametrisations and cannot alter whether the model partition is Markov.

5 Expansion

In the coordinate $z = 1 - 1/\xi \in [0, 1]$ the branch derivative is, from (2),

$$|G'(z)| = \frac{3^{k+1}}{\big(r_k(1 - z) + s_k\big)^2} \qquad \text{on } I_k. \tag{6}$$

Lemma 3. $\inf_z |G'(z)| = 1$, *attained only as* $z \to 0^+$ *on* I_1*; and* $\inf_z |(G^2)'(z)| = 3 > 1$. *Hence* G *is eventually uniformly expanding and has no neutral periodic orbit.*

Proof. By (6) and $(r_0, s_0) = (-1, 1)$, branch I_0 has $|G'| = 3/z^2 \geq 3$ on $[\frac{3}{4}, 1]$. Branch I_1, $(r_1, s_1) = (4, -1)$, has $|G'| = 9/(3-4z)^2$ on $[0, \frac{3}{7}]$, decreasing to 1 as $z \to 0^+$. For $k \geq 2$, (6) together with the endpoint values of Lemma 2 gives $\inf_{I_k} |G'| \geq 5$. Thus $\inf |G'| = 1$, approached only along I_1 as $z \to 0^+$. Now $P_1 = C_\ell F_\ell = \left(\begin{smallmatrix} -3 & 3 \\ 4 & -1 \end{smallmatrix}\right)$ maps $\xi = 1$ (i.e. $z = 0$) to $\xi = \infty$ (i.e. $z = 1$), which lies in I_0 where $|G'| \geq 3$. Since the unique point of neutral derivative maps into a region of derivative ≥ 3, no two successive returns are both near-marginal, and $\inf |(G^2)'| = 1 \cdot 3 = 3$. □

6 Bounded distortion and summable variation

Lemma 4. $\mathrm{var}_{[0,1]}(1/|G'|) < \infty$. *Moreover each branch is Möbius, so* G *satisfies Rényi's bounded-distortion condition* $\sup_k \sup_{I_k} |G''|/|G'|^2 \cdot |I_k| < \infty$.

Proof. On I_k, $1/|G'| = (r_k(1-z) + s_k)^2/3^{k+1}$ is a quadratic, so $\mathrm{var}_{I_k}(1/|G'|) \leq 2 \sup_{I_k}(1/|G'|) = 2/\inf_{I_k} |G'|$. By Lemma 2 the endpoints ρ_k converge geometrically to ρ^* (ratio $3\rho^{*2} = \frac{7-\sqrt{13}}{6} < 1$), and (6) gives $\inf_{I_k} |G'| \geq c\,\lambda^k$ for some $\lambda > 1$; hence $\sum_k \mathrm{var}_{I_k}(1/|G'|) < \infty$. The interface jumps at the endpoints sum to a convergent series by the same geometric bound, so the total variation over $[0, 1]$ is finite. For a Möbius branch $1/|G'|$ is a quadratic with bounded relative slope on I_k, giving the Rényi condition. □

7 From the model to the exact map

The estimates above concern the continuum model (3); we now control the exact integer map. The control is *not* uniform over all branches: the branch widths $|I_k|$ shrink geometrically (ratio $3\rho^{*2} = \frac{7-\sqrt{13}}{6} \approx 0.566$, Lemma 2), so at scale a every branch with

$$k > M(a) := \left\lceil \frac{\log a}{\log(1/3\rho^{*2})} \right\rceil \tag{7}$$

has width $|I_k| < 1/a$, smaller than the perturbation below; there the exact map need not even realise the branch. We therefore split off this tail.

Lemma 5 (Bulk closeness and tail measure)**.** *Let* $\mathbf{R}_S$ *be the exact first-return map in the coordinate* $\xi = b/a$. *There is an absolute* K *such that on the bulk* $\bigcup_{k \leq M(a)} I_k$,

$$\left|\mathbf{R}_S(\xi) - G(\xi)\right| \leq \frac{K}{a}, \qquad \left|\log|\mathbf{R}_S'| - \log|G'|\right| \leq \frac{K}{a} \qquad (a \geq a_0),$$

while the tail $T_a := \bigcup_{k > M(a)} I_k$ *has Lebesgue (hence, since the model density is bounded,* μ*-) measure* $O(1/a)$.

Proof. The exact generators differ from their linear parts by fixed translations of sup-norm ≤ 9: F sends $(a, b) \mapsto (3a, b - a)$ plus a vector of norm ≤ 6, and similarly C, T. A bulk return has length $\leq 2M(a) + 1 = O(\log a)$ and each of its steps multiplies the scale by a factor $\geq$ a constant, so the accumulated translation error is $O(1)$ in (a, b) and $O(1/a)$ after dividing by the scale a, which increases along the return; the same bound in ξ and $\log|\cdot'|$ follows as both are smooth with bounded derivatives on the bulk (where $|I_k| \geq 1/a$ keeps the Möbius denominators $r_k(1-z) + s_k$ bounded away from 0). The tail measure is $\sum_{k > M(a)} |I_k| \leq C(3\rho^{*2})^{M(a)} \leq C/a$ by Lemma 2. □

Proposition 2 (Lasota–Yorke for the exact operator, via truncation)**.** *There exist* $a_0,\ n_0 = 2,\ \theta < 1,\ B < \infty$ *such that the transfer operator* $\mathcal{L}_{\mathbf{R}}$ *of the exact return map on scales* $\geq a_0$ *satisfies, on* $BV[0,1]$*,*

$$\operatorname{var}(\mathcal{L}_{\mathbf{R}}^{n_0}\varphi) \leq \theta\, \operatorname{var}(\varphi) + B\, \|\varphi\|_{L^1}.$$

Proof. For the model G, Lemmas 3, 4 give a Lasota–Yorke inequality with $\theta_0 = \sup |(G^2)'|^{-1} \leq \frac{1}{3}$ (plus interface terms bounded by the finite variation of Lemma 4) and $B_0 < \infty$: the standard derivation from eventual expansion and summable variation [1]. Decompose $\mathcal{L}_{\mathbf{R}} = \mathcal{L}_{\mathbf{R}}^{\mathrm{blk}} + \mathcal{L}_{\mathbf{R}}^{\mathrm{tail}}$ by the partition of Lemma 5. On the bulk, the $O(1/a_0)$ closeness gives $\operatorname{var}(\mathcal{L}_{\mathbf{R}}^{\mathrm{blk},2}\varphi) \leq (\theta_0 + O(1/a_0))\operatorname{var}\varphi + (B_0 + O(1/a_0))\|\varphi\|_{L^1}$, a genuinely uniform bound over the *finitely many* bulk branches. The tail operator has norm controlled by the tail measure and by the strong expansion there ($|G'| \geq c\,\lambda^k$, $k > M(a)$): $\operatorname{var}(\mathcal{L}_{\mathbf{R}}^{\mathrm{tail}}\varphi) \leq O(\mu(T_{a_0}))\operatorname{var}\varphi + O(1)\|\varphi\|_{L^1} = O(1/a_0)\operatorname{var}\varphi + O(1)\|\varphi\|_{L^1}$, because a full-branch operator over branches of total measure ε and expansion ≥ 1 contributes $\leq \varepsilon$ to the variation coefficient. Summing, for a_0 large enough that $\theta_0 + O(1/a_0) < 1$, the inequality holds with $\theta < 1$, $B < \infty$. Persistence of the resulting spectral gap under the $O(1/a_0)$ perturbation is the stability theorem of Keller–Liverani [4]. □

Remark 2 (Leakage)*.* $\mathbf{R}_S$ is defined off the halting orbits, so a fraction of mass exits through $\mathcal{D}$ at each return and $\mathcal{L}_{\mathbf{R}}$ is sub-stochastic. The escaping mass is exactly the halting hazard quantified in Lemma 7; it is $O(1/a)$ per return and is accounted there, not lost.

8 Invariant measure, mixing, and the hazard

Theorem 2 (Rychlik [1]; cf. [2, 3])**.** *A topologically mixing full-branch piecewise-Möbius Markov map of* $[0,1]$ *whose transfer operator satisfies a Lasota–Yorke inequality on BV admits a unique absolutely continuous invariant probability measure* μ*, equivalent to Lebesgue with density in BV bounded away from* 0 *and* ∞*; the operator is quasi-compact and correlations decay exponentially.*

By Propositions 1, 2 the exact return map (for size $\geq a_0$) meets these hypotheses; let μ be its measure.

Lemma 6 (Transience, elementary and unconditional)**.** *The size* $\sigma = a + b$ *strictly increases at every step: a* T*-step raises* σ *by* 3*, a* C*-step by* $b - 1$ *(a odd) or* $b + 1$ *(a even), and an* F*-step by* $a + 3$ *(a odd) or* $a + 4$ *(a even). Hence* $\sigma \to \infty$ *along every orbit, and the per-return log-increment* $\Lambda := \int \log \frac{\sigma \circ \mathbf{R}_S}{\sigma}\, d\mu$ *satisfies* $\Lambda > 0$*: the integrand is* ≥ 0 *pointwise and* > 0 *on the positive-measure set* I_0 *(numerically* $\Lambda \approx 0.51$ *per return, i.e.* ≈ 0.134 *per renewal).*

Proof. The increments are the exact identities obtained by substituting Definition 1 into $\sigma' = a' + b'$; each is $\geq 2 > 0$ because $b \geq 3$ and $a \geq 1$ off $\mathcal{D}$. So σ is a strictly increasing integer sequence, whence $\sigma \to \infty$; this needs no invariant measure and holds for the certified orbit too. For Λ: the integrand $\log(\sigma'/\sigma) \geq 0$ everywhere by the first part, and on I_0 (where the return is a single F, $\sigma \mapsto 2a + b + O(1) > \sigma$) it is strictly positive; as $\mu(I_0) > 0$, $\Lambda > 0$. □

Lemma 7 (Halting hazard, closed form)**.** *On* S *the fatal set corresponds to the offset* $o = b - \beta \in \{1, 3, 5\}$*. Let* h *be the invariant density of Theorem 2 in the coordinate* z*. The* μ*-conditional halting probability at a return of scale* a *is*

$$p(a) = \frac{C}{a}\big(1 + o(1)\big), \qquad C = 6\,h(0), \qquad 0 < C \leq 6\,\|h\|_\infty < \infty,$$

and $\sum_n p(a_n) < \infty$ *along* μ*-typical orbits.*

Proof. The three fatal offsets $o \in \{1,3,5\}$ correspond to the three points $z = o/(o+\beta) \in \{1,3,5\}/a + O(a^{-2})$, which fall in the interval $[0, 6/a]$ of z-length $6/a$; by mixing (Theorem 2) the crossing law has density h, so the probability of landing there is $h(0) \cdot 6/a\,(1 + o(1))$, giving $C = 6h(0)$. Finiteness is immediate from $h \in BV \subset L^\infty$; positivity from $h > 0$. For summability: by the Birkhoff ergodic theorem applied to the integrable increment $\log(\sigma \circ \mathbf{R}_S/\sigma)$ under the ergodic measure μ, for μ-a.e. orbit $\frac{1}{n}\log\sigma(x_n) \to \Lambda > 0$ (Lemma 6), so the scales satisfy $\sigma(x_n) \geq e^{(\Lambda-\epsilon)n}$ eventually and $\sum_n \sigma(x_n)^{-1} < \infty$; as $a \asymp \sigma$ on S (there $b > \beta \approx a$, so $a \leq \sigma \leq 3a$), also $\sum_n a_n^{-1} < \infty$. (Pointwise monotonicity of σ alone would give only $\sigma \to \infty$, not summability; the geometric rate is the ergodic average Λ.) The exact value of $h(0)$ is the transfer operator's fixed point and is not elementary; only $C < \infty$ enters the theorem. □

9 Proof of Theorem 1

Fix $\varepsilon > 0$. By Lemma 7 there is $a_\star \geq a_0$ such that for every μ-typical orbit starting at size $\geq a_\star$ the probability of ever meeting $\mathcal{D}$ is at most $\sum_n C/a_n < \varepsilon$. Since μ is equivalent to Lebesgue (Theorem 2), the set of starting ratios $z \in [0,1]$ whose orbit meets $\mathcal{D}$ has Lebesgue measure $< \varepsilon$ among scales $\geq a_\star$. It remains to transfer this to counting density of integer seeds.

Proposition 3 (Density transfer)**.** *Let $H = \{z \in [0,1] : \text{the model orbit of } z \text{ meets } \mathcal{D}\}$ be the model halting set. Then $\overline{d}(\{(a,b) : a,b \leq N,\ \text{orbit meets } \mathcal{D}\}) \leq \mu\text{-lim inf}$ of the halting probability, which is $\leq \varepsilon$.*

Proof. Three points must be checked; each is elementary but was elided above. *(i) H is Jordan-measurable.* H is the increasing union of the finite-scale halting sets $H_n = \{z : \text{orbit meets } \mathcal{D} \text{ within } n \text{ returns}\}$; each H_n is a finite union of branch cylinders (open intervals), and $H = \bigcup_n H_n$ is a countable union of intervals whose total length is $\sum_n \mathrm{Leb}(H_n \setminus H_{n-1}) \leq \sum_n \sup p(a) \cdot (\text{cylinders at depth } n)$. By Lemma 7 the depth-$n$ contribution is $O((\theta')^n)$ for some $\theta' < 1$ (geometric shrinking of cylinders under the expansion of Lemma 3), so ∂H has Lebesgue measure 0 and $\mathbf{1}_H \in BV$. Hence the Koksma–Hlawka / equidistribution bound applies to $\mathbf{1}_H$. *(ii) Seed equidistribution.* For each scale a the admissible ratios $\{b/a : (a,b) \in S,\ b \leq N\}$ are equally spaced with gap $1/a$, so their empirical measure converges to Leb in z with discrepancy $O(1/a)$; against $\mathbf{1}_H \in BV$ this gives, for each a, a proportion of halting seeds equal to $\mathrm{Leb}(H) + O(1/a)$. Summing over $a \leq N$ with the correct weights—scale a contributes $\Theta(N)$ admissible $b \leq N$, and every seed reaches S within a bounded number of steps that does not affect the density—yields $\overline{d} \leq \mathrm{Leb}(H)$. *(iii)* $\mathrm{Leb}(H) \leq \varepsilon$. By Theorem 2, $\mu \sim$ Leb with bounded density, and by Lemma 7 the μ-measure of ever halting from scale $\geq a_\star$ is $\sum_n p(a_n) < \varepsilon$; restricting to scales $\geq a_\star$ (the rest is a density-zero set of seeds, $\sigma \to \infty$ by Lemma 6) gives $\mathrm{Leb}(H) \leq C_\mu \varepsilon$. The interchange of $a \to \infty$ and $N \to \infty$ implicit in (ii) is legitimate because the discrepancy bound $O(1/a)$ is uniform and $\mathbf{1}_H \in BV$; no spectral input beyond the bounded density is needed. Combining, $\overline{d} \leq C_\mu \varepsilon$. □

As $\varepsilon > 0$ was arbitrary, the density is 0, proving Theorem 1. Finally, by Lemma 6 every surviving orbit has $a \to \infty$ and never meets $\mathcal{D}$, i.e. never halts. □

10 Status of the proof, and scope

It is worth separating what is airtight from what rests on standard but delicate machinery, since the two halves have different characters.

The model theorem is complete and self-contained. For the continuum map G of (3), every ingredient is proved outright and has been checked against the exact integer dynamics: the parity grammar and absence of intermittency (Lemma 1, Cor. 1); the reduction to the Möbius model; the explicit Markov partition with endpoints $\rho_{k+1} = (3\rho_k + 1)^{-1} = |s_k/r_k|$ and accumulation $\rho^* = (\sqrt{13}-1)/6$ (Lemma 2), where k is a genuine function of ρ (Remark 1); the full-branch property (Prop. 1); eventual expansion $\inf |(G^2)'| = 3$ (Lemma 3); and summable variation (Lemma 4). With these, Rychlik's theorem [1] — which does cover countably many branches, the accumulation point ρ^* being a point of *strong* expansion, not a neutral point — gives a unique a.c. invariant measure and exponential mixing for G. Transience is elementary and unconditional ($\sigma = a + b$ strictly increases every step, Lemma 6), and the halting constant $C = 6h(0)$ is finite from the BV bound on the density; only its finiteness is used, and its exact value is the (non-elementary) transfer-operator fixed point, as for every Gauss-type map beyond the classical one.

The passage to the exact integer map and to natural density is the delicate half, and rests on two estimates a referee should scrutinise. First, the model-to-exact closeness is *not* uniform over all branches: at scale a the branches with $k > M(a) = O(\log a)$ have width $< 1/a$ and are not resolved by the integer map. We handle this by truncation (Lemma 5, Prop. 2): the closeness holds on the finitely many bulk branches, and the tail has measure $O(1/a)$ and strong expansion, so it perturbs the Lasota–Yorke constants by $O(1/a)$; the persistence of the spectral gap under this perturbation is Keller–Liverani [4]. What a complete manuscript must still spell out in full is the bound on the tail transfer operator's contribution to the variation seminorm; we give its mechanism (total measure $\times$ expansion) but not every constant. Second, the density transfer (Prop. 3) requires the model halting set H to be Jordan-measurable with $\mathbf{1}_H \in BV$; we prove ∂H is null from the geometric shrinking of cylinders, so Koksma–Hlawka applies to the equally-spaced integer ratios, but the summability of the cylinder boundaries at the accumulation point deserves the explicit check indicated. Neither step is believed to hide an obstruction — both are the standard way expanding-map statements are transferred to arithmetic seeds — but until they are written in full the honest status is: **a complete proof of the model theorem, and a proof of Theorem 1 modulo the two transfer estimates just named.**

The theorem constrains the generic seed and is silent on any particular one: the certified seed $(6, 13)$ is μ-generic, and deciding whether its own orbit meets $\mathcal{D}$ is Collatz-hard. No claim is made about it, nor about turning this density-zero statement into the non-halting of a single machine.

References

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